© 2021 ASHRAE, Inc.

NAHA, Japan

WMO: **479360**

Lat: **26.2072N** Lon: **127.6872E** Elev: **50** StdP: **100.73** Time Zone: **9.00 (E09)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	12.0	12.9	2.5	4.5	13.7	3.5	4.9	14.2	12.4	14.5	11.5	15.3	6.1	0	0.484	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	4.2	32.3	26.4	31.8	26.3	31.3	26.2	27.5	30.4	27.2	30.1	26.9	29.9	5.4	90	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	26.7	22.4	29.2	26.4	22.0	29.1	26.1	21.6	29.0	87.8	30.3	86.4	30.1	85.1	30.0	28.8	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	12.5	10.8	9.5	DB	10.3	33.6	1.2	1.1	9.5	34.4	8.8	35.0	8.1	35.7	7.3	36.5	(4)
(5)				WB	6.6	27.9	1.0	0.6	5.9	28.3	5.3	28.6	4.8	28.9	4.0	29.3	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	23.5	17.2	17.6	19.1	21.6	24.4	27.3	29.3	29.2	28.1	25.8	22.6	19.1	(6)
(7)			DBStd	4.77	2.34	2.62	2.58	2.28	1.94	1.83	1.02	1.05	1.31	1.74	2.06	2.36	(7)
(8)			HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)			HDD18.3	138	50	42	24	3	0	0	0	0	0	0	1	19	(9)
(10)			CDD10.0	4917	224	213	281	349	445	519	598	594	544	489	379	282	(10)
(11)			CDD18.3	2014	16	22	46	102	187	269	340	336	294	231	130	43	(11)
(12)			CDH23.3	18202	4	12	48	268	1064	2752	4273	4178	3321	1804	441	37	(12)
(13)			CDH26.7	5900	0	0	1	12	131	803	1820	1731	1084	298	21	1	(13)
(14)	Wind		WSAvg	5.2	5.3	5.2	5.2	5.1	4.8	5.4	5.3	5.2	5.3	5.4	5.3	5.3	(14)
(15)	Precipitation		PrecAvg	2067	113	111	151	164	245	262	176	252	213	164	119	112	(15)
(16)			PrecMax	3326	275	336	420	525	637	861	510	674	1096	689	367	260	(16)
(17)			PrecMin	964	22	20	28	34	13	20	5	22	23	7	4	1	(17)
(18)			PrecStd	494	55	72	88	104	146	157	147	148	179	129	77	65	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		24.0	24.7	26.0	27.9	30.0	32.1	33.1	33.2	32.3	30.9	28.4	25.6	(19)
(20)			MCWB		20.1	20.8	22.2	24.2	25.4	26.8	26.3	26.6	26.2	25.4	24.0	21.5	(20)
(21)		2%	DB		22.7	23.5	24.6	26.8	29.0	31.2	32.4	32.4	31.6	29.9	27.2	24.4	(21)
(22)			MCWB		19.0	20.1	21.0	23.4	25.0	26.6	26.3	26.4	26.0	25.1	22.9	20.3	(22)
(23)		5%	DB		21.8	22.5	23.6	25.8	28.2	30.6	31.9	31.7	31.0	29.0	26.4	23.4	(23)
(24)			MCWB		18.1	19.4	20.3	22.5	24.6	26.5	26.3	26.3	25.7	24.3	22.3	19.4	(24)
(25)		10%	DB		20.9	21.6	22.8	24.9	27.2	30.0	31.4	31.2	30.3	28.2	25.6	22.6	(25)
(26)			MCWB		17.2	18.5	19.8	22.0	24.0	26.3	26.2	26.2	25.5	23.9	21.8	18.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		21.4	22.1	22.9	25.0	26.5	27.7	28.0	27.7	27.2	26.4	25.1	22.8	(27)
(28)			MCDB		23.3	23.6	24.6	26.9	28.6	30.3	30.6	30.8	30.5	29.2	27.1	24.5	(28)
(29)		2%	WB		20.0	21.2	22.1	24.2	25.8	27.3	27.5	27.4	26.8	25.8	24.1	21.6	(29)
(30)			MCDB		21.9	23.0	23.9	26.0	27.8	29.7	30.4	30.5	30.0	28.6	26.3	23.5	(30)
(31)		5%	WB		18.9	20.1	21.3	23.5	25.2	27.0	27.2	27.1	26.5	25.4	23.4	20.4	(31)
(32)			MCDB		21.1	21.9	23.2	25.3	27.3	29.4	30.2	30.2	29.6	28.1	25.7	22.7	(32)
(33)		10%	WB		17.8	19.0	20.3	22.7	24.7	26.7	26.8	26.7	26.2	24.9	22.6	19.4	(33)
(34)			MCDB		20.3	21.2	22.3	24.5	26.8	29.1	29.9	29.9	29.2	27.6	25.2	22.0	(34)
(35)	Mean Daily Temperature Range		MDBR		4.2	4.3	4.5	4.5	4.2	3.9	4.2	4.1	4.1	4.0	4.0	4.1	(35)
(36)		5% DB	MCDBR		5.3	5.2	5.3	4.9	4.7	4.2	4.5	4.6	4.6	4.5	4.5	4.9	(36)
(37)			MCWBR		3.8	3.7	3.6	2.9	2.4	1.6	1.7	1.7	1.9	2.2	2.7	3.4	(37)
(38)		5% WB	MCDBR		4.6	4.7	4.8	4.6	4.3	4.0	4.1	4.2	4.2	4.1	4.0	4.3	(38)
(39)			MCWBR		4.1	4.3	4.3	3.3	2.8	1.9	1.8	1.8	1.9	2.1	2.9	3.8	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.457	0.486	0.560	0.568	0.511	0.470	0.470	0.488	0.483	0.482	0.465	0.457	(40)
(41)		taud			2.135	2.074	1.902	1.942	2.167	2.348	2.340	2.278	2.261	2.205	2.210	2.143	(41)
(42)		Ebn at Noon			777	787	750	757	799	827	826	811	807	782	766	757	(42)
(43)		Edh at Noon			136	155	194	191	152	126	127	135	134	135	126	130	(43)
(44)	All-Sky Solar Radiation	RadAvg			2.44	3.08	3.80	4.43	4.95	5.48	6.51	5.96	5.23	4.12	3.01	2.46	(44)
(45)		RadStd			0.29	0.46	0.36	0.45	0.64	0.50	0.53	0.41	0.40	0.36	0.28	0.24	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+0.28	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	+0.41	+0.53	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page